GROUNDING WITH LOG PROBABILITIES

OpenAI's API includes a feature that enables models to return log probabilities for each token generated, providing valuable insights into how confident the model is about its answers. To utilize this feature effectively, you need to pass additional configuration options when initializing the API request. This allows the model to also evaluate the likelihood of each token it produces, which can be particularly useful for assessing the reliability of the information provided.

Here is a quick example of how to set this up when you create an instance of the `ChatOpenAI` class:

```
const openAI = await new ChatOpenAI({
    model: 'gpt-3.5-turbo',
    apiKey: OpenAIKey,
    logprobs: true,
    topLogprobs: 3,
  });
```

Two additional options when we instantiate the OpenAI client are

- `logprobs`—Enables the return of log probabilities for each token
- `top_logprobs`—Specifies that the top three probable tokens should be returned along with their probabilities

A response might look like the following when querying about a specific topic, such as "What are the benefits of exercise?":

```
"logprobs": {
        "tokens": ["Regular", "exercise", "improves",
"physical", "health", ",", "boosts", "mental",
"well-being", ",", "and", "enhances", "overall",
"quality", "of", "life", "."],
        "token_logprobs": [-0.5, -0.7, -0.6, -0.8,
-0.9, -1.0, -0.4, -0.5, -0.6, -0.7, -0.8, -0.9, -1.0,
-1.1, -1.2, -1.3],
        "top_logprobs": [
          {"token": "Regular", "logprob": -0.5},
          {"token": "Consistent", "logprob": -0.6},
          {"token": "Daily", "logprob": -0.7},
          {"token": "exercise", "logprob": -0.7},
          {"token": "activity", "logprob": -0.8},
          {"token": "workout", "logprob": -0.9}
        ]
      }
```

NOTE Not all LLMs offer support for log probabilities in their public API methods. While this feature is available in OpenAI's API, the implementation and response formats may differ between platforms. Carefully review the documentation for each API to understand how log probabilities are handled and how they can be utilized effectively in their applications.